Problem Statement:

Write code to simulate requests coming from clients and distribute them among the servers using the load balancing algorithms.

Setup

In [1]:
```
servers = ["S1", "S2", "S3"]
requests = list(range(1, 11))  # 10 requests
```

Round Robin Algorithm

In [2]:
```
print("=== Round Robin ===")

rr_allocation = {}
index = 0

for req in requests:
    server = servers[index % len(servers)]
    rr_allocation[req] = server
    print(f"Request {req} → {server}")
    index += 1
```

```
=== Round Robin ===
Request 1 → S1
Request 2 → S2
Request 3 → S3
Request 4 → S1
Request 5 → S2
Request 6 → S3
Request 7 → S1
Request 8 → S2
Request 9 → S3
Request 10 → S1
```

Least Connections Algorithms

In [3]:
```
print("\n=== Least Connections ===")

# Initialize connection count
connections = {server: 0 for server in servers}
lc_allocation = {}

for req in requests:
    # Find server with least connections
    server = min(connections, key=connections.get)

    lc_allocation[req] = server
    connections[server] += 1

    print(f"Request {req} → {server} | Load: {connections}")
```

```
=== Least Connections ===
Request 1 → S1 | Load: {'S1': 1, 'S2': 0, 'S3': 0}
Request 2 → S2 | Load: {'S1': 1, 'S2': 1, 'S3': 0}
Request 3 → S3 | Load: {'S1': 1, 'S2': 1, 'S3': 1}
Request 4 → S1 | Load: {'S1': 2, 'S2': 1, 'S3': 1}
Request 5 → S2 | Load: {'S1': 2, 'S2': 2, 'S3': 1}
Request 6 → S3 | Load: {'S1': 2, 'S2': 2, 'S3': 2}
Request 7 → S1 | Load: {'S1': 3, 'S2': 2, 'S3': 2}
Request 8 → S2 | Load: {'S1': 3, 'S2': 3, 'S3': 2}
Request 9 → S3 | Load: {'S1': 3, 'S2': 3, 'S3': 3}
Request 10 → S1 | Load: {'S1': 4, 'S2': 3, 'S3': 3}
```

Request Completion Simulation

In [4]:
```
import random

connections = {server: 0 for server in servers}

for req in requests:
    server = min(connections, key=connections.get)
    connections[server] += 1

    print(f"Request {req} → {server} | Load: {connections}")

    # Randomly complete a request
    finished_server = random.choice(servers)
```

```
    if connections[finished_server] > 0:
        connections[finished_server] -= 1
print(f"Completed from {finished_server}")
```

```
Request 1 → S1 | Load: {'S1': 1, 'S2': 0, 'S3': 0}
Request 2 → S1 | Load: {'S1': 1, 'S2': 0, 'S3': 0}
Request 3 → S2 | Load: {'S1': 1, 'S2': 1, 'S3': 0}
Request 4 → S1 | Load: {'S1': 1, 'S2': 1, 'S3': 0}
Request 5 → S3 | Load: {'S1': 1, 'S2': 1, 'S3': 1}
Request 6 → S1 | Load: {'S1': 1, 'S2': 1, 'S3': 1}
Request 7 → S1 | Load: {'S1': 1, 'S2': 1, 'S3': 1}
Request 8 → S1 | Load: {'S1': 1, 'S2': 1, 'S3': 1}
Request 9 → S1 | Load: {'S1': 1, 'S2': 1, 'S3': 1}
Request 10 → S2 | Load: {'S1': 1, 'S2': 1, 'S3': 1}
Completed from S2
```